20261013 Quiz

for (MATH301) Modern Algebra I

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Exercise 1.7.18. Let H be a group acting on a set A. Prove that the relation $\sim$ on A defined by

$$a \sim b \quad \Longleftrightarrow \quad a = h \cdot b \text{ for some } h \in H$$

is an equivalence relation. For each $x \in A$, the equivalence class of x under $\sim$ is called the orbit of x under the action of H. The orbits under the action of H partition the set A.

Solution. For every $a \in A$,

$$a = 1 \cdot a,$$

so $a \sim a$. If $a \sim b$, then $a = h \cdot b$ for some $h \in H$, and hence

$$b = h^{-1} \cdot a,$$

so $b \sim a$. Finally, if $a \sim b$ and $b \sim c$, say $a = h \cdot b$ and $b = k \cdot c$, then

$$a = h \cdot (k \cdot c) = (hk) \cdot c,$$

so $a \sim c$. Thus $\sim$ is an equivalence relation.

For $x \in A$, its equivalence class is

$$[x] = \{h \cdot x \mid h \in H\} = H \cdot x,$$

which is precisely the orbit of x. Hence the orbits partition A.

□

Exercise 2.2.4. For each of S_3, D_4, and Q_8, compute the centralizers of each element and find the center of each group. Does Lagrange's Theorem (Exercise 1.7.19) simplify your work?

Notation. Following MATH000, D_n denotes the dihedral group of symmetries of a regular n-gon, so $|D_n| = 2n$. Thus D_4 here is denoted by D_8 in Dummit–Foote.

Solution. In S_3, the identity commutes with every element. A transposition commutes only with itself and the identity, while the two 3-cycles commute with each other. Thus

x	$C_{S_3}(x)$
1	S_3
$(12), (13), (23)$	$\langle x \rangle$
$(123), (132)$	$\langle (123) \rangle$

and therefore

$$Z(S_3) = \{1\}.$$

Write

$$D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle.$$

Then r^2 is central, rotations commute with one another, and the reflections split into two commuting pairs. Hence

x	$C_{D_4}(x)$
$1, r^2$	D_4
r, r^3	$\langle r \rangle$
s, sr^2	$\langle s, r^2 \rangle$
sr, sr^3	$\langle sr, r^2 \rangle$

so

$$Z(D_4) = \{1, r^2\}.$$

In $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, the elements ± 1 are central, and each of $\pm i$, $\pm j$, and $\pm k$ commutes precisely with the corresponding cyclic subgroup of order 4. Thus

x	$C_{Q_8}(x)$
± 1	Q_8
$\pm i$	$\langle i \rangle$
$\pm j$	$\langle j \rangle$
$\pm k$	$\langle k \rangle$

and

$$Z(Q_8) = \{\pm 1\}.$$

Lagrange's Theorem simplifies these computations because $C_G(x) \leq G$ and $\langle x \rangle \leq C_G(x)$, so the possible orders of a centralizer are strongly restricted once one finds an element that does not commute with x.

□

Exercise 2.2.6. Let H be a subgroup of the group G.

(a) Show that $H \leq N_G(H)$. Give an example to show that this is not necessarily true if H is not a subgroup.

(b) Show that $H \leq C_G(H)$ if and only if H is abelian.

Solution.

(a) Let $h \in H$. Since H is a subgroup, $hxh^{-1} \in H$ for every $x \in H$, so

$$hHh^{-1} \subseteq H.$$

Applying the same argument to h^{-1} gives the reverse inclusion. Hence $hHh^{-1} = H$, so $h \in N_G(H)$. Therefore

$$H \leq N_G(H).$$

The conclusion can fail when H is only a subset. In S_3, let

$$H = \{(12), (13)\}.$$

Then $(12) \in H$, but

$$(12)H(12)^{-1} = \{(12), (23)\} \neq H.$$

Thus $(12) \notin N_{S_3}(H)$, so $H \not\subseteq N_{S_3}(H)$.

(b) If $H \leq C_G(H)$, then every element of H commutes with every element of H, so H is abelian. Conversely, if H is abelian, then every $h \in H$ commutes with every element of H, hence $h \in C_G(H)$. Therefore

$$H \leq C_G(H) \quad \Longleftrightarrow \quad H \text{ is abelian.}$$

□

Exercise 2.4.3. Prove that if H is an abelian subgroup of a group G, then $\langle H, Z(G)\rangle$ is abelian. Give an explicit example of an abelian subgroup H of a group G such that $\langle H, C_G(H)\rangle$ is not abelian.

Solution. Every element of $Z(G)$ commutes with every element of G, and the elements of H commute with one another. Hence every word in elements of $H \cup Z(G)$ can be rearranged into the form hz with $h \in H$ and $z \in Z(G)$. For $h_1, h_2 \in H$ and $z_1, z_2 \in Z(G)$,

$$(h_1z_1)(h_2z_2) = h_1h_2z_1z_2 = h_2h_1z_2z_1 = (h_2z_2)(h_1z_1).$$

Thus $\langle H, Z(G)\rangle$ is abelian.

For a counterexample, let $G = D_4$ and $H = \langle r^2\rangle = Z(D_4)$. Then H is abelian and

$$C_{D_4}(H) = D_4.$$

Therefore

$$\langle H, C_{D_4}(H)\rangle = D_4,$$

which is not abelian.

□

Exercise 2.4.7. Prove that the subgroup of S_4 generated by (12) and (13)(24) is isomorphic to D_4, the dihedral group of order 8.

Solution. Let

$$a = (12), \qquad b = (13)(24), \qquad r = ab, \qquad s = a.$$

Both a and b have order 2, and $r = ab$ is a 4-cycle, so $|r| = 4$. Also

$$s^2 = 1$$

and

$$srs = a(ab)a = ba = (ab)^{-1} = r^{-1}.$$

Thus r and s satisfy the defining relations of D_4.

Moreover,

$$\langle a, b\rangle = \langle r, s\rangle,$$

since $s = a$ and $b = sr$. The relations above reduce every element of $\langle r, s\rangle$ to one of

$$1, r, r^2, r^3, s, sr, sr^2, sr^3.$$

Hence $|\langle r, s\rangle| \leq 8$. Since $\langle r\rangle$ has order 4 and the transposition s is not in $\langle r\rangle$, the cosets $\langle r\rangle$ and $s\langle r\rangle$ are distinct, so $|\langle r, s\rangle| \geq 8$. Therefore $|\langle r, s\rangle| = 8$. The assignment from the usual generators of D_4 to r and s defines a surjective homomorphism

$$D_4 \to \langle r, s\rangle.$$

Since both groups have order 8, this homomorphism is an isomorphism.

□

Exercise 2.4.14. A group H is called finitely generated if there is a finite set A such that $H = \langle A \rangle$.

(a) Prove that every finite group is finitely generated.

(b) Prove that $\mathbb{Z}$ is finitely generated.

(c) Prove that every finitely generated subgroup of the additive group $\mathbb{Q}$ is cyclic.

(d) Prove that $\mathbb{Q}$ is not finitely generated.

Hint. For part (c), if H is a finitely generated subgroup of $\mathbb{Q}$, show that $H \leq \langle 1/k \rangle$, where k is the product of all denominators appearing in a set of generators for H.

Solution.

(a) If H is finite, then $H = \langle H \rangle$, and H itself is a finite generating set. Hence every finite group is finitely generated.

(b) Under addition,
$$\mathbb{Z} = \langle 1 \rangle,$$
so $\mathbb{Z}$ is finitely generated.

(c) Suppose
$$H = \left\langle \frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n} \right\rangle \leq \mathbb{Q},$$
where each $b_i > 0$, and let
$$k = b_1 b_2 \cdots b_n.$$
For each i,
$$\frac{a_i}{b_i} = a_i \frac{k}{b_i} \cdot \frac{1}{k} \in \left\langle \frac{1}{k} \right\rangle.$$
Hence
$$H \leq \left\langle \frac{1}{k} \right\rangle.$$
Since every subgroup of a cyclic group is cyclic, H is cyclic.

(d) If $\mathbb{Q}$ were finitely generated, then part (c) would imply that $\mathbb{Q}$ is cyclic. But if $q \neq 0$, then $q/2 \notin \langle q \rangle$, while $\langle 0 \rangle = \{0\}$. Thus $\mathbb{Q}$ is not cyclic and therefore is not finitely generated.

□

Exercise 2.4.15. Exhibit a proper subgroup of $\mathbb{Q}$ which is not cyclic.

Solution. Let
$$H = \left\{ \frac{m}{2^k} \,\middle|\, m \in \mathbb{Z},\ k \in \mathbb{N}_0 \right\}.$$
If $m/2^k, n/2^\ell \in H$, then
$$\frac{m}{2^k} - \frac{n}{2^\ell} = \frac{2^\ell m - 2^k n}{2^{k+\ell}} \in H,$$
so $H \leq \mathbb{Q}$. It is proper because $1/3 \notin H$.

Suppose $H = \langle q \rangle$ for some $q \in H$. If $q = 0$, then $\langle q \rangle \neq H$. If $q = m/2^k \neq 0$, then
$$\frac{1}{2^{k+1}} \in H$$
cannot be an integer multiple of q, since
$$n \frac{m}{2^k} = \frac{1}{2^{k+1}}$$
would imply $2nm = 1$. Hence H is not cyclic.

□

Exercise 2.5.3. Show that the subgroup $\langle s, r^2 \rangle$ of D_4 is isomorphic to V_4.

Solution. Since $r^4 = s^2 = 1$ and $sr^2 = r^2 s$, we have

$$\langle s, r^2 \rangle = \{1, r^2, s, sr^2\}.$$

Define

$$\varphi \colon (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}) \to \langle s, r^2 \rangle, \qquad \varphi([i]_2, [j]_2) = r^{2i} s^j.$$

Because r^2 and s commute and both have order 2, φ is a homomorphism. Its four values are precisely

$$1, r^2, s, sr^2,$$

so it is bijective. Since $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}) \cong V_4$,

$$\langle s, r^2 \rangle \cong V_4.$$

□

Exercise 2.5.9. Draw the lattices of subgroups of the following groups:

(a) $\mathbb{Z}/16\mathbb{Z}$,

(b) $\mathbb{Z}/24\mathbb{Z}$,

(c) $\mathbb{Z}/48\mathbb{Z}$.

Hint. See Exercise 2.3.6.

Solution. For each divisor d of n, the cyclic group $\mathbb{Z}/n\mathbb{Z}$ has a unique subgroup of order d, namely

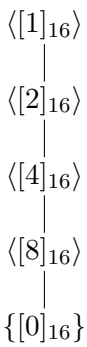
$$H_d = \langle [n/d]_n \rangle,$$

and

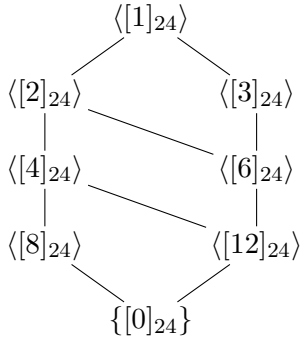
$$H_d \leq H_e \quad \Longleftrightarrow \quad d \mid e.$$

Thus the subgroup lattices are the divisor lattices of 16, 24, and 48.

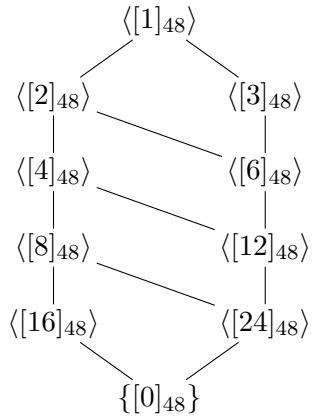
(a)



(b)



(c)



□

Exercise 3.1.1. Let G and H be groups. Let $\varphi\colon G \to H$ be a homomorphism and let E be a subgroup of H. Prove that $\varphi^{-1}(E) \leq G$, i.e., the preimage or pullback of a subgroup under a homomorphism is a subgroup. If $E \trianglelefteq H$, prove that $\varphi^{-1}(E) \trianglelefteq G$. Deduce that $\ker\varphi \trianglelefteq G$.

Solution. Since $\varphi(1_G) = 1_H \in E$, the set $\varphi^{-1}(E)$ is nonempty. If $x, y \in \varphi^{-1}(E)$, then

$$\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} \in E,$$

so $xy^{-1} \in \varphi^{-1}(E)$. By the Subgroup Criterion,

$$\varphi^{-1}(E) \leq G.$$

Now suppose $E \trianglelefteq H$. If $g \in G$ and $x \in \varphi^{-1}(E)$, then

$$\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g)^{-1} \in E,$$

because E is normal in H. Hence

$$\varphi^{-1}(E) \trianglelefteq G.$$

Taking $E = \{1_H\} \trianglelefteq H$ gives

$$\ker\varphi = \varphi^{-1}(\{1_H\}) \trianglelefteq G.$$

□

Exercise 3.1.2. Let $\varphi\colon G \to H$ be a homomorphism of groups with kernel K, and let $a, b \in \varphi(G)$. Let $X \in G/K$ be the fiber above a and let Y be the fiber above b, i.e.,

$$X = \varphi^{-1}(a), \qquad Y = \varphi^{-1}(b).$$

Fix an element $u \in X$, so $\varphi(u) = a$. Prove that if $XY = Z$ in the quotient group G/K and w is any member of Z, then there is some $y \in Y$ such that $uy = w$.

Hint. Show that $u^{-1}w \in Y$.

Solution. By the description of fibers in §3.1, $X = uK$. Choose $v \in Y$, so $Y = vK$. Since

$$Z = XY = (uK)(vK) = uvK,$$

any $w \in Z$ has the form

$$w = uvk$$

for some $k \in K$. Set

$$y = u^{-1}w = vk.$$

Since $vk \in vK = Y$, we have $y \in Y$, and by construction

$$uy = w.$$

□

Exercise 3.1.7. Define $\pi\colon \mathbb{R}^2 \to \mathbb{R}$ by

$$\pi((x, y)) = x + y.$$

Prove that π is a surjective homomorphism and describe the kernel and fibers of π geometrically.

Solution. For $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$\begin{aligned}\pi((x_1, y_1) + (x_2, y_2)) &= \pi((x_1 + x_2, y_1 + y_2)) \\ &= x_1 + x_2 + y_1 + y_2 \\ &= \pi((x_1, y_1)) + \pi((x_2, y_2)).\end{aligned}$$

Thus π is a homomorphism. It is surjective because for every $c \in \mathbb{R}$,

$$\pi((c, 0)) = c.$$

The kernel is

$$\ker \pi = \{(x, y) \in \mathbb{R}^2 \mid x + y = 0\} = \{(t, -t) \mid t \in \mathbb{R}\},$$

the line through the origin with slope -1. More generally, the fiber above $c \in \mathbb{R}$ is

$$\pi^{-1}(c) = \{(x, y) \in \mathbb{R}^2 \mid x + y = c\},$$

which is the line of slope -1 with y-intercept c. Hence the fibers are parallel affine lines, each a coset of $\ker \pi$.

□